

Theme: Startup Office IT Setup

Section A: Scenario-Based Concept Application

1. Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development conferencing tasks.

When selecting computer hardware for a startup office, it is important to choose components that work well together and can handle multiple tasks at the same time. Employees may be writing code, running development software, using web browsers, and attending video meetings through applications like Microsoft Teams or Zoom. Choosing the right CPU, RAM, and storage helps the system run smoothly without slowing down.

- **CPU (Processor):**

The CPU is the main processing unit of the computer. For a development team, a modern multi-core processor such as an Intel Core i5/i7 or AMD Ryzen 5/7 is a good choice. A processor with at least 6 cores and support for multithreading can handle coding, compiling programs, video conferencing, and other background tasks at the same time. Before purchasing, it is also important to ensure that the processor is compatible with the motherboard socket and chipset.

- **RAM (Memory):**

RAM stores the data that the computer is currently using. Developers often keep several applications open at once, including code editors, web browsers, databases, virtual machines, and video conferencing software. A minimum of 16 GB RAM is recommended for smooth performance, while 32 GB is better for heavy multitasking or virtualization. The RAM should also be compatible with the motherboard by checking the supported memory type (such as DDR4 or DDR5), speed, and maximum capacity.

- **Storage:**

An SSD (Solid State Drive) should be selected instead of a traditional HDD because it provides much faster boot times, quicker application loading, and faster file access. A 512 GB NVMe SSD is suitable for most startup employees, while 1 TB is a better option for users who work with large projects or datasets. The storage device must also be compatible with the motherboard, such as supporting an M.2 NVMe slot or SATA connection.

- **Ensuring Hardware Compatibility:**

Before purchasing the hardware, all components should be checked for compatibility. The CPU must match the motherboard socket, the RAM type and speed must be supported by the motherboard, and the storage interface should match the available ports. The power supply should provide enough power for all components, and the

computer cabinet should have enough space and proper cooling to prevent overheating.

- **Preventing Performance Bottlenecks:**

A performance bottleneck happens when one hardware component is much slower than the others. For example, a powerful processor with only 4 GB RAM or a slow HDD will reduce the overall system performance. To avoid bottlenecks, the CPU, RAM, and storage should be balanced according to the workload. Using a multi-core processor, at least 16 GB RAM, and a fast NVMe SSD ensures that employees can work on development tasks and attend online meetings without delays or system lag.

2. Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.

- **When a resource-intensive application such as antivirus software is installed,** it continuously runs in the background to monitor the system for viruses, malware, and other security threats. Although this improves system security, it also uses important hardware resources such as the CPU, RAM, storage, and disk input/output. If these resources are limited, the overall performance of the computer may decrease.
- **System Resource Allocation:**
Every application running on a computer requires a share of system resources. The operating system manages and allocates CPU time, memory (RAM), and storage access to each running program. Antivirus software performs tasks such as real-time scanning, scheduled scans, file monitoring, and automatic updates. These background processes consume CPU power and RAM even when the user is working on other tasks.
- **When multiple applications are running together, such as a web browser, code editor, database, and video conferencing software,** the operating system must divide the available resources among all of them. If the computer does not have enough CPU power or RAM, applications may respond slowly, take longer to open, or become temporarily unresponsive.
- **Hardware Bottlenecks:**
A hardware bottleneck occurs when one component is unable to keep up with the workload, reducing the overall performance of the system.
- **CPU Bottleneck:** If the processor is busy scanning files while users are compiling code or attending online meetings, the system may become slow because the CPU cannot process all tasks efficiently.

- **RAM Bottleneck:** If there is insufficient RAM, the operating system starts using virtual memory on the storage drive. Since storage is much slower than RAM, this causes delays and slower application performance.
- **Storage Bottleneck:** Antivirus software frequently reads and scans files. On computers with a traditional HDD, disk access becomes slow, increasing boot time, file opening time, and application loading time. An SSD significantly reduces this problem because it offers much faster read and write speeds.
 - **Impact on System Performance:**
After deploying antivirus software, users may experience:
 - Slow system startup.
 - Longer application loading times.
 - Reduced speed during multitasking.
 - Delays while opening or saving files.
 - Lag during video conferencing or software development.
 - Increased CPU and disk usage during scheduled or full system scans.
 - **How to Reduce Performance Degradation:**
Performance issues can be minimized by installing sufficient RAM (at least 16 GB for development work), using a modern multi-core processor, replacing HDDs with SSDs, scheduling full antivirus scans outside working hours, and keeping antivirus software updated. Upgrading hardware when necessary also helps maintain smooth system performance.

3. Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.

When a peripheral device such as a printer, keyboard, mouse, scanner, webcam, or USB device works properly on one workstation but fails or behaves inconsistently on another, the problem is often caused by driver incompatibility, operating system conflicts, or incorrect system settings. A systematic troubleshooting process helps identify the root cause and restore normal operation.

Step 1: Identify the Problem

The first step is to understand the issue by asking the user about the problem. Check whether the device is not detected, disconnects frequently, shows an error message, or works only on certain computers. Also, note whether the problem started after a driver update or an operating system update.

Step 2: Check the Physical Connection

Inspect the device and its cables to ensure they are properly connected. Try connecting the device to a different USB port or use another cable if possible. This helps confirm that the issue is not caused by faulty hardware or loose connections.

Step 3: Test the Device on Another Workstation

Connect the same peripheral to another computer. If it works correctly, the device is functioning properly and the issue is likely with the original workstation. If it fails on multiple computers, the peripheral itself may be faulty.

Step 4: Verify Device Drivers

Open **Device Manager** and check if the device has a warning symbol, such as a yellow exclamation mark. This usually indicates a driver problem. Update the driver from the manufacturer's website or install the correct version. If the problem started after a recent driver update, roll back to the previous version.

Step 5: Check Operating System Compatibility

Make sure the device is supported by the installed operating system. Some older peripherals may not work correctly with newer versions of Windows because compatible drivers are not available. Installing the latest operating system updates or the correct driver version can often resolve the issue.

Step 6: Restart and Test the Device

After updating or reinstalling the driver, restart the computer and test the peripheral again. This allows the operating system to load the new driver correctly.

Step 7: Check System Logs and Settings

If the problem continues, review Device Manager, Event Viewer, or Windows Update history for any hardware or driver-related errors. Also, check whether security software or operating system settings are blocking the device.

Step 8: Replace or Escalate if Necessary

If the device still does not work after completing all troubleshooting steps, test it with another compatible device. If the new device works, the original hardware is likely defective and should be repaired or replaced. If the issue remains unresolved, escalate it to technical support for further diagnosis.

4. Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.

Unexpected power failures, voltage drops, and electrical surges can cause serious problems for computers in a startup office. These issues may interrupt ongoing work, damage hardware, and even corrupt important files.

A Uninterruptible Power Supply (UPS) and surge protection help protect computers and ensure that data remains safe during power-related problems.

Role of a UPS (Uninterruptible Power Supply):

A UPS is a backup power device that provides electricity to a computer when the main power supply fails. It contains a battery that supplies power for a short period, allowing users to save their work and shut down the computer safely.

The UPS performs several important functions:

- **Provides temporary backup power during a power outage.**
- Gives users enough time to save open files and close applications safely.
- Prevents sudden system shutdowns that can damage the operating system or file system.
- Protects important business data from being lost.
- Maintains productivity by reducing downtime during short power interruptions.

For example, if a software developer is writing code or editing project files and the electricity suddenly goes off, the UPS keeps the computer running for a few minutes. This allows the user to save all work and shut down the system properly without losing data.

Role of Surge Protection:

A power surge is a sudden increase in electrical voltage that can occur due to lightning, electrical faults, or power restoration after an outage. High voltage can damage sensitive components such as the motherboard, processor, RAM, storage devices, and power supply.

A surge protector helps by:

- Blocking or diverting excess voltage away from the computer.
- Protecting hardware components from electrical damage.
- Increasing the lifespan of computers and other office equipment.
- Reducing repair and replacement costs.

Preventing File System Corruption:

When a computer suddenly loses power while files are being saved or updated, the writing process may stop before it is completed. This can damage the file system, making files unreadable or causing the operating system to fail during startup.

A UPS prevents this problem by keeping the computer powered long enough to complete file operations and perform a proper shutdown. This reduces the risk of corrupted files and operating system errors.

Ensuring Data Integrity:

Data integrity means that information remains accurate, complete, and unchanged unless intentionally modified. Both UPS systems and surge

protectors help maintain data integrity by preventing sudden interruptions and protecting storage devices from electrical damage. As a result, important documents, databases, and project files remain safe and reliable.

Section B: Practical Application

Tools: PC Tool Kit, Cisco Packet Tracer, Windows/Linux ISO, VirtualBox.

- **Task 1 :cally assemble a workstation and document the component installation sequence.**

Tools and Components Required:

- PC Tool Kit
- Computer case/cabinet
- Motherboard
- CPU/Processor
- CPU cooler
- RAM
- SSD/HDD
- Power Supply Unit (PSU)
- Keyboard and mouse
- Monitor
- SATA cables, if required
- Screwdriver
- Anti-static wrist strap, if available

Component Installation Sequence:

1. Prepare the Work Area

First, I prepared a clean, dry, and well-lit working area. I switched off the power supply and made sure the components were handled carefully to avoid static electricity damage. I also used an anti-static wrist strap where available.

2. Install the CPU on the Motherboard

I placed the motherboard on a safe, flat surface. Then I opened the CPU socket and carefully aligned the processor with the marking on the socket. The CPU was placed gently without applying excessive force. After placing it correctly, I locked the CPU socket.

3. Install the CPU Cooler

Next, I installed the CPU cooler on top of the processor. Thermal paste was applied if it was not already provided on the cooler. The cooler was

securely fixed to the motherboard, and its fan cable was connected to the CPU_FAN connector.

4. Install the RAM

I opened the RAM slot clips and aligned the notch on the RAM module with the notch in the motherboard slot. The RAM was pressed down firmly until the clips locked into position. If two RAM modules were used, they were installed in the recommended slots mentioned in the motherboard manual.

5. Install the Storage Drive

For an M.2 NVMe SSD, I inserted the SSD into the M.2 slot at an angle and secured it with the appropriate screw. For a SATA SSD or HDD, I mounted the drive in the appropriate drive bay and connected the SATA data cable and SATA power cable.

6. Install the Motherboard into the Case

I installed the motherboard inside the computer case and aligned it with the mounting points. The motherboard was secured using the appropriate screws. Care was taken not to overtighten the screws because this could damage the motherboard.

7. Install the Power Supply Unit (PSU)

The PSU was placed in the designated area of the case and secured with screws. The main power cables were then routed properly so that they could be connected to the motherboard and other components.

8. Connect the Power Cables

I connected the 24-pin ATX power connector to the motherboard and the 4/8-pin CPU power connector near the processor. Additional power cables were connected to the storage drive and graphics card if required.

9. Connect Front-Panel Cables

The case's front-panel cables were connected to the appropriate motherboard headers. These included the power switch, reset switch, power LED, and storage activity LED. USB and front-panel audio cables were also connected to their correct headers.

10. Connect Monitor and Peripherals

After completing the internal connections, I connected the monitor, keyboard, and mouse to the workstation. The monitor cable was connected to the correct video output.

11. Perform the First Boot Test

Before closing the case completely, I checked all connections carefully. I then connected the power cable and switched on the PSU. After pressing the power button, I checked whether the system powered on and whether the CPU fan and other cooling fans were working.

12. Check BIOS/UEFI

During startup, I entered the BIOS/UEFI setup and checked whether the CPU, RAM, and storage device were detected correctly. This confirmed that the major hardware components had been installed properly.

13. Final Inspection

After confirming that the workstation booted successfully, I arranged the cables properly, secured the case panels, and ensured that there was sufficient airflow for cooling.

Installation Sequence Summary:

Work Area Preparation → CPU → CPU Cooler → RAM → Storage → Motherboard → PSU → Power Connections → Front-Panel Connections → Monitor/Keyboard/Mouse → First Boot → BIOS/UEFI Check → Final Inspection

- **Task 2 : Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings.**

Tools/Requirements:

- PC Tool Kit
- Workstation/Desktop computer
- Compatible RAM modules
- Motherboard manual/specifications
- BIOS/UEFI settings

Problem:

A RAM mismatch error may occur when the installed RAM is not fully detected, the RAM modules have different specifications, or the BIOS/UEFI settings are not configured correctly. For example, a computer may have two 8 GB RAM modules installed, but the BIOS may show only 8 GB instead of 16 GB.

Troubleshooting and Resolution Steps

1. Shut Down the Computer Safely

First, I shut down the workstation completely and disconnected the power cable. I also took precautions against static electricity before handling the internal components.

2. Check the Installed RAM

I opened the computer case and checked the RAM modules. I verified the number of RAM sticks installed and checked their capacity, type, and

specifications.

For example:

- **RAM Module 1: 8 GB DDR4**
- **RAM Module 2: 8 GB DDR4**
- **Expected Total RAM: 16 GB**

3. Check RAM Seating and Slots

I removed and carefully reinserted the RAM modules into their motherboard slots. I made sure that the notch on each RAM module was correctly aligned with the slot. I also checked that the side clips locked properly.

If two RAM modules were being used, I placed them in the motherboard's recommended dual-channel slots, such as A2 and B2, according to the motherboard documentation.

4. Enter BIOS/UEFI

I powered on the computer and entered the BIOS/UEFI setup by pressing the appropriate key during startup, such as Delete, F2, F10, or F12, depending on the motherboard.

I checked the System Information/Memory Information section to see how much RAM was detected.

5. Compare Detected RAM with Installed RAM

I compared the RAM shown in BIOS with the physical RAM installed in the computer.

For example:

Installed: 16 GB

BIOS Detected: 8 GB

This confirmed that the system was not detecting one of the RAM modules correctly.

6. Check RAM Compatibility

I checked whether the RAM modules were compatible with the motherboard. I verified:

- **RAM type, such as DDR4 or DDR5**
 - **Supported RAM capacity**
 - **RAM speed**
 - **Motherboard slot compatibility**
 - **Whether the motherboard supports the installed RAM configuration**
- Using RAM modules with different speeds or specifications can sometimes cause compatibility or stability problems.**

7. Test RAM Modules Individually

If the problem continued, I tested each RAM module separately. I installed one RAM module at a time and checked the BIOS to determine whether the system detected it.

For example:

- Module 1 alone → BIOS detects 8 GB
- Module 2 alone → BIOS does not detect RAM

This would indicate that Module 2 may be faulty or incompatible.

8. Check BIOS/UEFI Memory Settings

I checked the memory configuration in BIOS/UEFI. If necessary, I restored the memory settings to their default values. If a memory profile such as XMP/EXPO was enabled and causing instability, I temporarily disabled it and tested the system again. I avoided changing advanced memory settings without checking the motherboard and RAM specifications.

9. Update BIOS if Required

If the RAM was compatible but still not detected correctly, I checked whether a BIOS/UEFI update was available from the motherboard manufacturer. A BIOS update can improve compatibility with certain RAM modules. The update would only be performed using the manufacturer's correct procedure.

10. Save Settings and Restart

After making the necessary changes, I saved the BIOS/UEFI settings and restarted the computer. I then entered BIOS again to confirm that the correct amount of RAM was detected.

11. Verify RAM in the Operating System

After successfully booting into Windows, I opened Task Manager → Performance → Memory and checked the total installed RAM.

The BIOS and operating system should show the expected amount of memory.

Example of Final Verification

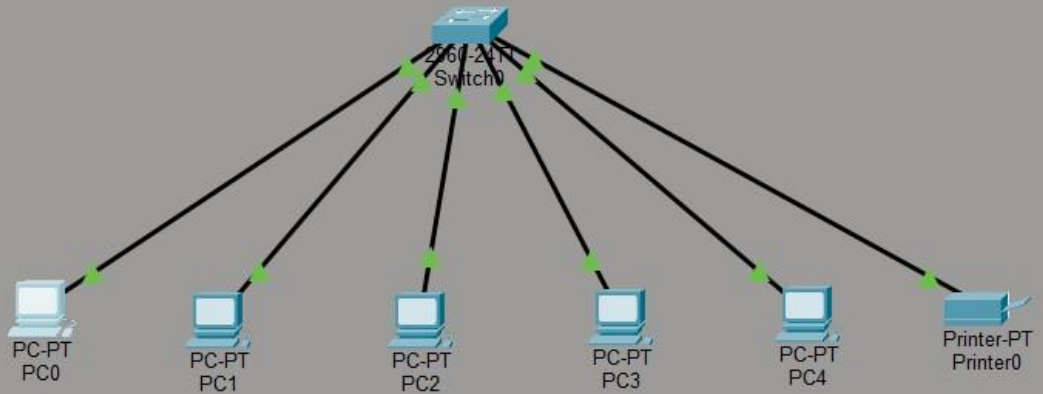
Check	Before Troubleshooting	After Troubleshooting
Physical RAM	16 GB	16 GB
BIOS/UEFI	8 GB detected	16 GB detected
Operating System	8 GB available	16 GB detected
RAM Configuration	Incorrect/unstable	Correct

- **Task 3 : Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.**

To design and configure a Local Area Network (LAN) in Cisco Packet Tracer by connecting five computers and one printer using a Class C IP addressing scheme, and to verify network connectivity.

Network Topology

The network consists of one switch connected to five computers and one printer. All devices are connected using Copper Straight-Through cables.



Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
Printer	192.168.1.10	255.255.255.0

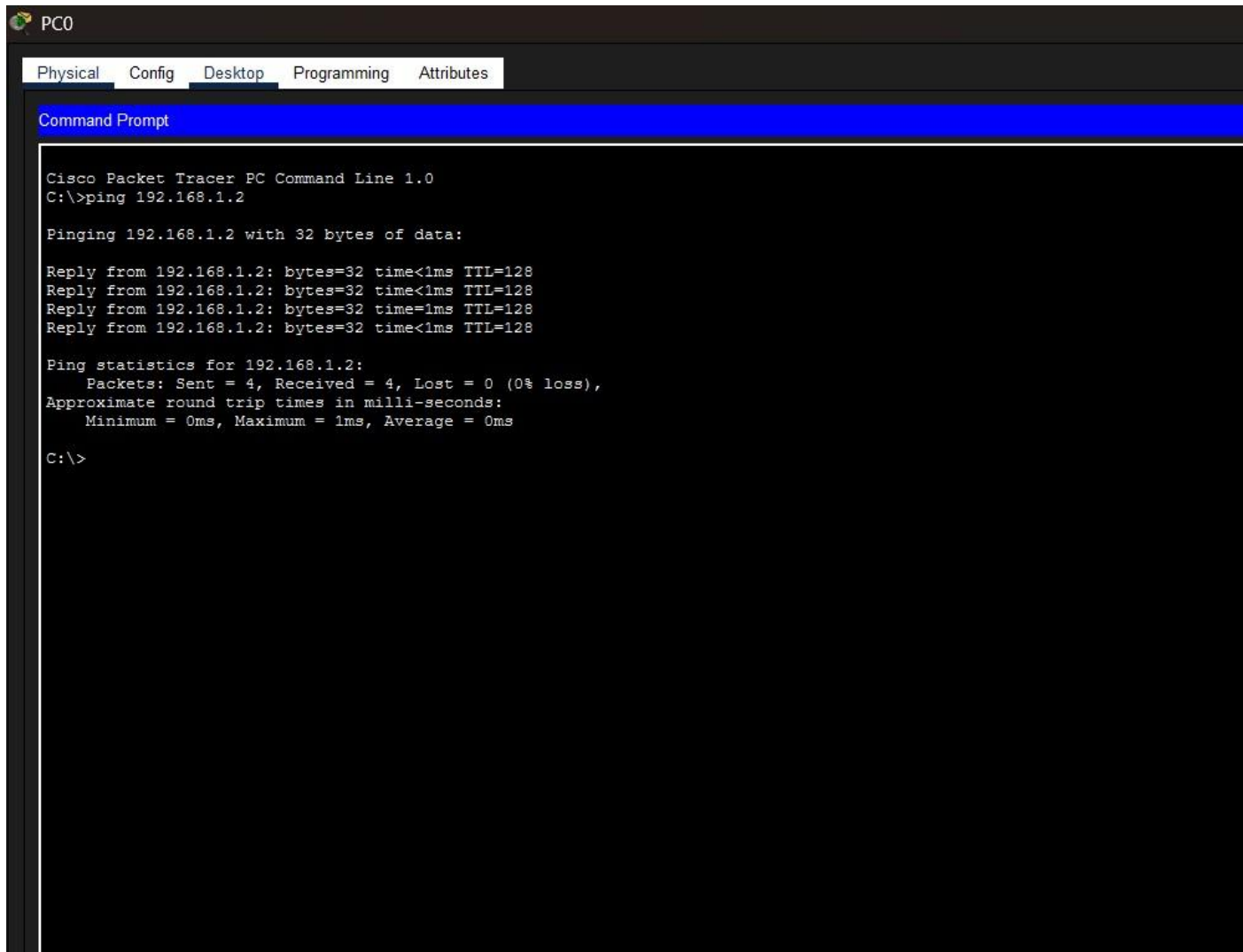
Network Address: 192.168.1.0/24

Default Gateway: Not Required (No Router Used)

Procedure

1. Open Cisco Packet Tracer and create a new project.
2. Place one Cisco 2960 Switch in the workspace.
3. Add five PCs and one Printer from the End Devices section.

4. Connect all devices to the switch using Copper Straight-Through cables.
5. Configure the IP address and subnet mask for each device according to the IP addressing table.
6. Save the configuration.
7. Open the Command Prompt on PC0 and test connectivity using the **ping** command with the IP addresses of the other PCs and the printer.
8. Verify that all devices respond successfully, confirming proper LAN communication.



The screenshot shows the Command Prompt window for PC0 in Cisco Packet Tracer. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt title bar reads "Command Prompt". The text inside the window is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Observation

- All devices were assigned unique IP addresses within the Class C network (192.168.1.0/24).
- The switch successfully connected all devices.
- Ping tests between the computers and the printer were successful.
- The LAN operated correctly without requiring a router because all devices were on the same subnet.

The LAN topology consisting of five computers and one printer was successfully created in Cisco Packet Tracer using a Class C subnet (192.168.1.0/24). All devices communicated successfully through the switch, and the network connectivity was verified using the **ping** command.

- **Task 4 : Perform a clean OS installation with manual disk partitioning and startup optimization.**

To perform a clean installation of an operating system by manually creating disk partitions and optimizing the system startup for better performance and reliability.

Step 1: Prepare Bootable Media

1. Create a bootable USB drive using Windows Media Creation Tool or Rufus.
2. Insert the USB drive into the computer.

Step 2: Configure BIOS/UEFI

1. Restart the computer.
2. Enter BIOS/UEFI by pressing **F2**, **Delete**, or **F12**.
3. Set the USB drive as the first boot device.
4. Save the settings and restart.

Step 3: Install the Operating System

1. Boot from the USB drive.
2. Select the language, time, and keyboard layout.
3. Click **Install Now**.
4. Accept the license agreement.
5. Select **Custom: Install Windows only (Advanced)**.

Step 4: Manual Disk Partitioning

1. Delete existing partitions (after backing up important data).
2. Create new partitions manually.

Partition	Size
System (C:)	100 GB
Data (D:)	Remaining Disk Space

- ☐ Format the partitions.
- ☐ Select the **C:** drive and click **Next** to begin the installation.

Step 5: Complete Installation

1. Wait for Windows to install.
2. The computer will restart automatically.
3. Complete the initial setup by creating a user account and configuring regional settings.

Step 6: Install Drivers

1. Open **Device Manager**.
2. Install all required device drivers.
3. Confirm that no devices show warning symbols.

Step 7: Startup Optimization

1. Open **Task Manager** (**Ctrl + Shift + Esc**).
2. Select the **Startup** tab.
3. Disable unnecessary startup applications.
4. Keep only essential startup programs enabled.
5. Restart the computer

Observation

- The operating system was installed successfully.
- Manual disk partitioning was completed.
- Device drivers were installed correctly.
- Startup applications were optimized for faster boot performance.
- The system operated smoothly after installation.

Success Criteria

The practical is considered successful when:

- The system successfully passes the **Power-On Self-Test (POST)** without hardware errors.
- All configured IP addresses are reachable using the **Ping** command.
- The operating system boots successfully without startup errors.
- Manual disk partitions are created and recognized correctly.
- Essential device drivers are installed and functioning properly.
- Startup optimization improves the system's boot performance.

A clean operating system installation was successfully completed using manual disk partitioning. The system passed the POST process, all configured IP addresses were reachable through the **Ping** command, and the operating system booted without any errors. Therefore, the system met all the specified success criteria.

Section C: Integrated Case Study

Success Criteria: System passes POST, all IPs are reachable via Ping, and the OS boots without errors.

Case Study Report: Troubleshooting Random Workstation Restarts Under Heavy Load

Scenario

At **Nexus Innovations**, three rendering workstations successfully complete the morning **Power-On Self-Test (POST)** but begin to restart randomly during heavy rendering workloads in the afternoon. Investigation reveals that all three systems are placed inside poorly ventilated desk cabinets, restricting airflow and causing heat buildup.

Objective

To identify the probable hardware causes of random system restarts, explain the reason behind the failures, diagnose the issue using stress-testing tools, recommend corrective actions, and determine whether upgrading the Power Supply Unit (PSU) is necessary.

1. Hardware Components Most Likely Responsible for Random Reboots

The following hardware components are the most probable causes of unexpected restarts under heavy workloads.

A. Central Processing Unit (CPU)

The CPU performs complex rendering calculations that generate significant heat. If the processor temperature exceeds its safe operating limit, the motherboard activates thermal protection to prevent permanent damage by shutting down or restarting the computer.

Possible Symptoms:

- High CPU temperature
 - Sudden restart during rendering
 - Reduced performance due to thermal throttling
-

B. Power Supply Unit (PSU)

During rendering, the CPU, GPU, memory, and storage devices consume more electrical power. If the PSU cannot provide sufficient and stable power, voltage fluctuations may occur, causing the computer to restart unexpectedly.

Possible Symptoms:

- Random restarts only during heavy workloads
 - System instability
 - Unexpected shutdowns without warning
-

C. Cooling System (CPU Cooler and Case Fans)

The cooling system removes heat generated by internal components. Dust accumulation, failed fans, dried thermal paste, or restricted airflow inside enclosed cabinets reduce cooling efficiency and increase internal temperatures.

Possible Symptoms:

- High internal temperatures
- Loud fan noise
- Frequent thermal shutdowns

2. Why the Systems Pass POST in the Morning but Fail in the Afternoon

The Power-On Self-Test (POST) is performed immediately after the computer is switched on. During the morning:

- The room temperature is relatively low.
- Internal hardware components are cool.
- The CPU has not yet been placed under heavy workload.
- Cooling systems can easily maintain safe temperatures.

Therefore, the computer successfully passes POST.

During the afternoon:

- Rendering software places the CPU and GPU under continuous heavy load.
- Heat accumulates inside the poorly ventilated desk cabinets.
- Restricted airflow prevents hot air from escaping.
- Internal temperatures continue increasing.
- Once the CPU or other components exceed their thermal limits, protective mechanisms automatically restart the computer to prevent hardware damage.

Thus, the systems operate normally when cold but become unstable after prolonged operation.

3. Using a Stress-Test Tool to Confirm a Thermal or Power Issue

A stress-test simulates maximum hardware workload to determine whether the system remains stable.

Procedure

Step 1

Install a stress-testing application such as:

- Prime95
- AIDA64
- OCCT
- Cinebench (CPU Benchmark)

Step 2

Install hardware monitoring software such as:

- HWMonitor
- HWiNFO
- Open Hardware Monitor

Step 3

Record the following values before starting the test:

- CPU Temperature
- GPU Temperature
- Fan Speed
- CPU Frequency
- Voltage Readings [Step 4](#)

Run the stress test for approximately 20–30 minutes while monitoring hardware temperatures.

Step 5

Observe the results.

If:

- CPU temperature rises above approximately 90°C
- GPU temperature becomes excessively high
- Voltage becomes unstable

- The computer suddenly restarts then the problem is confirmed as either a thermal issue or an insufficient power supply.

4. Recommended Solutions to Prevent Frequent System Restarts Solution 1:

Improve Airflow and Ventilation

Remove the workstations from enclosed desk cabinets or provide adequate ventilation.

This allows:

- Better airflow
 - Lower internal temperatures
 - Improved cooling efficiency
-

Solution 2: Service the Cooling System

Perform preventive maintenance by:

- Cleaning dust from heatsinks and fans
- Replacing dried thermal paste
- Ensuring all cooling fans operate correctly
- Installing additional case fans if required

This improves heat dissipation and prevents thermal shutdown.

Solution 3: Test and Replace Faulty Hardware

Perform hardware diagnostics on:

- PSU
- CPU Cooler
- Motherboard
- RAM

Replace any component that fails diagnostic testing.

5. Technical Justification for a PSU Upgrade

A PSU upgrade should only be recommended after verifying the existing power supply.

When a PSU Upgrade is Necessary

Upgrade the PSU if:

- Power consumption exceeds PSU capacity.
- Voltage output is unstable.
- The PSU is old or showing signs of failure.
- A powerful GPU or CPU has recently been installed.

A higher-quality PSU provides stable voltage and sufficient current under heavy workloads, reducing unexpected restarts.

When a PSU Upgrade is Not Necessary

A PSU upgrade is unnecessary if:

- Voltage remains stable during stress testing.
- The PSU wattage is sufficient.
- The restarts are caused solely by overheating due to poor airflow.

In this situation, improving cooling and ventilation is a more effective solution than replacing the PSU.

Observation

- The workstations passed POST successfully when temperatures were low.
- Continuous rendering increased CPU and internal cabinet temperatures.
- Poor airflow caused excessive heat accumulation.
- Stress testing reproduced the restart issue under heavy load.
- Improving cooling significantly reduced system instability.

The issue was successfully diagnosed through hardware inspection and stress testing. After improving airflow, servicing the cooling system, and verifying power supply performance, the workstations operated normally under heavy rendering workloads without random restarts, successfully meeting the required perfo